

KAI BIJLANI

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Education

University of California, San Diego

June 2027

B.S. in Data + Computer Science, B.S. in Mathematics

La Jolla, CA

- GPA: 3.88 / 4.00
- Courses: DSA/Advanced Algorithms, RDBMS, Distributed & Scalable Systems, Probabilistic ML, Linear Algebra, Discrete Math

Technical Skills

Languages: Python, R, Java, JavaScript, TypeScript, C, C++, SQL (MySQL, PostgreSQL), HTML, CSS, Bash, GitLab CLI
Tools/Platforms: Databricks, AWS (EC2, S3, SageMaker), MLFlow, Apache Spark/Hadoop, Docker, Redis, Kubernetes, Jenkins
Libraries/Frameworks: PyTorch, TensorFlow, Flask, Django, React, Node.js, OpenCV, Pandas, Sklearn, Pytest, Express.js

Experience

Talmo Lab | Salk Institute

February 2025 – Present

Computer Vision Research Intern

La Jolla, CA

- Utilized PyTorch and time-series animal behavioral data to optimize 3D pose estimation + phenotyping conclusions with SLEAP
- Used Kalman filtering, 3D CNN-based triangulation, graph NNs, and part affinity fields (PAFs) for multi-animal segmentation.
- Containerized builds and tracked artifacts with Git/Actions, Docker and Kubernetes, using custom Marimo notebooks for GUI.

Data Science Student Society (DS3)

May 2024 – Present

Vice President of Software, Director of Projects

La Jolla, CA

- Built and deployed collaborative filtering and deep learning models for Amazon using TensorFlow and AWS SageMaker.
- Developed Tesla's semantic segmentation model with UNet, OpenCV, PySpark, and deployed real-time inference on AWS EC2.
- Full stack software developer for the DS3 external and internal website, work in Node.js + React and a school-developed VM.
- Developed MongoDB database, managing REST APIs with Express.js to optimize query times and reduce JSON format reliance.

Rivian

June 2025 – September 2025

Software Engineering Intern

Irvine, CA

- Developed R1 failure agents with Databricks/Spark, parsing Kinesis streamed JSON logs in an iterative preprocessing pipeline.
- Used BERT/self-tuned embedders to represent high-dimensional ECU data, passed to clustering-classifier semi-supervised agent.
- Built a Databricks Asset Bundle to deploy + version-control as a containerized job, logged results to MLFlow/Unity Catalog.

UCSD - College of Computing (HDSI)

July 2024 - February 2025

Machine Learning Researcher

La Jolla, CA

- Trained LSTM and Transformer models using PyTorch, Pandas, and AWS EC2, with experiments tracked via MLflow.
- Built ETL pipelines on 100+ GB of data using PySpark and SQL, and sequenced features with Scikit-Learn, Matplotlib/Seaborn.
- Developed continuous testing workflows and artifact versioning systems using Docker, Git/Actions and Jenkins, caching model and data iterations with 85+% accuracy using automated Statistical Process Control (SPC) techniques.

Lavner

June 2024 – September 2024

Software Engineering Intern

San Francisco, CA

- Managed MySQL databases with Redis caching, achieving sub-100 ms access times for high-demand 150+ GB application data.
- Built backend solutions using Python, Flask, and Twilio APIs, optimizing DB management with Redis and Docker containers.
- Automated CI/CD builds and deployments with Git/Actions and AWS EC2, ensuring high scalability and system reliability.

CarPro-S

March 2024 – June 2024

Founding Engineer

La Jolla, CA

- Built an AI-powered driving kit with an RPI, camera, pressure sensors and backend live-time CV model, to detect unsafe driving.
- Optimized facial landmark detection using TensorFlow, CV2, and SVMs, improving gaze tracking with hyperparameter tuning.
- Designed notification system with Twilio's SMS API in Flask to alert authorities/trigger warnings, reliability tested with Pytest.

Projects

Machine Learning Under the Hood | *Python, Numpy, FastAPI, TypeScript, MongoDB, React*

September 2025 – Present

- Productionalized a fullstack, FastAPI-based app hosted with Render and built with React + Node.js, used in live-time.
- Built derivations of key machine-learning algorithms such as Gaussian Mixture Modeling and PCA, tracked states via React.

Keys of Parkinson's | *React, TypeScript, MongoDB, D3.js, Machine Learning, NLP*

October 2024 - January 2025

- Built a full-stack app with ingestion, preprocessing, and interactive analysis on early Parkinson's biomarkers from typing patterns.
- Used NLP (Naive-Bayes, logistic regression, cross-val) to simulate keystroke patterns, optimized with suitable data structures.

Multimodal Audio Classification with Signal Processing | *Python, PyTorch, TorchAudio*

October 2023 – January 2024

- Utilized Mel Spectrograms to convert time-series data to a logarithmic scale, sampling at 48kHz to avoid outliers.
- Built a music classification program with RNN-CNN joint architecture and double-layered LSTM, achieving 80% accuracy.

Awards and Honors

3x AIME Qualifier, Teaching Assistant (DSA), UCSD Undergraduate Research Scholar, National Merit Scholar, Published Author